TVI Innovations -- enterprise logical architecture (target design)

Internet / SaaS
Microsoft 365, cloud IoT hub, public DNS

SASE / ZTNA PoP
identity-aware proxy, SWG, CASB, FWaaS

TLS / ZTNA

Remote workforce
ZTNA client + MFA

ISP-A (AS 65100)
Broadband Internet
Transport A

peering

ISP-B (AS 65200)
MPLS / business fibre
Transport B

overlay A

overlay B

DEMARC

DEMARC

Hanoi R&D Campus (AS 65020)

1,500 users -> 3,500 | 10.20.0.0/16 | 2001:db8:a020::/48

HAN-EDG1
NGFW (ZBF) + SD-WAN

HAN-EDG2
NGFW (ZBF) + SD-WAN

Access
router-on-a-stick, HSRP

R&D engineers
VLAN 20

AI lab GPU cluster
10.20.64.0/18

Guests
VLAN 60

Ho Chi Minh City HQ (AS 65010)

2,000 users -> 5,000 | 10.10.0.0/16 | 2001:db8:a010::/48 | data centre

HCM-EDG1
NGFW (ZBF) + SD-WAN

HCM-EDG2
NGFW (ZBF) + SD-WAN

LACP

CORE1
L3 + HSRP

CORE2
L3 + HSRP

border

DC fabric: EVPN-VXLAN
spine-leaf, OSPF underlay, iBGP EVPN

Spine1

Spine2

Leaf1

Leaf2

Access
802.1X, DHCP snooping, DAI

Wi-Fi 6 APs
CAPWAP -> WLC

Executives
VLAN 10

WLC / DHCP
DNS / NTP

Da Nang Logistics Hub (AS 65030)

500 users + 10k IoT | 10.30.0.0/16 | 2001:db8:a030::/48

DAD-EDG1
NGFW (ZBF) + SD-WAN

DAD-EDG2
NGFW (ZBF) + SD-WAN

Access
router-on-a-stick, HSRP

Wi-Fi 6/7 APs
low-latency roaming

AGVs, scanners
VLAN 50, isolated

Logistics staff
VLAN 30

Dashed blue/orange: GRE-over-IPsec overlay, full mesh on each transport, eBGP between sites. Each site breaks out to the Internet locally (no backhaul through HQ).